

Problem Set

11.

1) uGV $\exists w' \in W$. So, $T \in L(w, v)$
by $Tw = \langle w, w' \rangle u$.

To find: $T^\#$

$$\text{So, } \langle Tw, v \rangle = \langle w, T^\#v \rangle$$

$$\therefore \langle \langle w, w' \rangle u, v \rangle$$

$$= \langle \overline{w, w'} \rangle \langle u, v \rangle.$$

$$\text{Now, } \langle \overline{w, w'} \rangle = \langle w', w \rangle$$

$$\therefore \langle w', w \rangle \langle u, v \rangle$$

$$\Rightarrow \langle w, \langle u, v \rangle w' \rangle.$$

$$\boxed{T^\#v = \langle u, v \rangle w'}$$

2). To prove: every tne operator on V has unique tne square root.

Proof:

Let $T \in L(V)$ be tne, i.e. $\langle Tv, v \rangle \geq 0$
 $\forall v$.

$\therefore \exists$ a unique tne operator S st.

$$S^2 = T,$$

As T is tne, it's self-adjoint.

$$\Delta Te_i = \lambda_i e_i, \quad \lambda_i \geq 0.$$

$$\begin{aligned}
 \text{So, } S e_i &= (\sqrt{\lambda_i}) e_i \\
 \therefore S^2 e_i &= S(S e_i) = S(\sqrt{\lambda_i} e_i) \\
 &= \sqrt{\lambda_i} (S e_i) \\
 &= \sqrt{\lambda_i} S e_i \\
 &= (\sqrt{\lambda_i})^2 e_i = \lambda_i e_i \\
 &= T e_i
 \end{aligned}$$

$$\therefore S^2 = T$$

Now, for $v = \sum c_i e_i$

$$\begin{aligned}
 \langle S v, v \rangle &= \langle \sum c_i \sqrt{\lambda_i} e_i, \sum c_i e_i \rangle \\
 &= \sum \sqrt{\lambda_i} |c_i|^2 \geq 0
 \end{aligned}$$

$\therefore S$ is positive.

Let R be a true operator

$$\text{s.t. } R^2 = T$$

Since T is diagonalizable in $\mathbb{R}^n$

R must preserve eigenspaces.

$$\therefore Re_i = \mu_i e_i.$$

$$\therefore R^2 e_i = \mu_i^2 e_i = \lambda_i e_i = \lambda_i e_i$$

$$\therefore \mu_i^2 = \lambda_i$$

R is true so $\mu_i \geq 0$.

$$\therefore \mu_i = \sqrt{\lambda_i}$$

$$\therefore R = S$$

and hence S is unique

So, every true operator has a
UNIQUE true square root.

3)

a) To prove: if V is a real inner product space,

$$4 \cdot \langle u, v \rangle = \|u+v\|^2 - \|u-v\|^2$$

b) To prove: if V is a complex inner product space,

$$4 \langle u, v \rangle = \|u+v\|^2 - \|u-v\|^2 + \|u+iv\|^2 - \|u-iv\|^2$$

c) if V is finite dimensional, $S, T \in GL(V)$
 prove that: $ST = I$ iff $TS = I$.

Proof:

a) $\|u\|^2 = \langle u, u \rangle$

$$\|u+v\|^2 = \langle u+v, u+v \rangle = \langle u, u \rangle$$

$$+ \langle u, v \rangle + \langle v, u \rangle + \langle v, v \rangle$$

$\therefore$ it's a real space,

$$\langle u, v \rangle = \langle v, u \rangle$$

$$\therefore \|u+v\|^2 = \|u\|^2 + 2\langle u, v \rangle + \|v\|^2$$

$$\|u-v\|^2 = \|u\|^2 - 2\langle u, v \rangle + \|v\|^2$$

$$\therefore \|u+v\|^2 - \|u-v\|^2 = 4\langle u, v \rangle$$

b) It's a complex inner product space,

$$\langle v, u \rangle = \overline{\langle u, v \rangle}$$

$$\therefore \|u+v\|^2 = \|u\|^2 + \|v\|^2 + \langle u, v \rangle + \langle v, u \rangle$$

$$\|u-v\|^2 = \|u\|^2 + \|v\|^2 - \langle u, v \rangle - \langle v, u \rangle$$

$$\therefore \|u+v\|^2 - \|u-v\|^2 = 2(\langle u, v \rangle + \langle v, u \rangle)$$

$$= 4\operatorname{Re}(\langle u, v \rangle)$$

Now,

$$\|u+iv\|^2 = \|u\|^2 + \|v\|^2 + i\langle u, v \rangle - i\langle v, u \rangle$$

$$\|u-iv\|^2 = \|u\|^2 + \|v\|^2 + i\langle v, u \rangle - i\langle u, v \rangle$$

$$\therefore \|u+iv\|^2 - \|u-iv\|^2 = 2i(\langle u, v \rangle - \langle v, u \rangle) = 4\operatorname{Im}\langle u, v \rangle$$

$$\therefore \|u+iv\|^2 - \|u-iv\|^2 + i\|u+iv\|^2 - i\|u-iv\|^2$$

$$= 4(\operatorname{Re}\langle u, v \rangle + i(\operatorname{Im}\langle u, v \rangle))$$

$$= \boxed{4\langle u, v \rangle}$$

c). Prove:

$$ST = I \quad \Leftrightarrow \quad TS = I$$

Assume: $ST = I$

Now, if $Tv = 0$,

$$v = Iv = ST(v) = S(0) = 0$$

so, nullspace of $S = \{0\}$.

Now, as it's finite dimensional,

if S is injective it means
 S is surjective.

$\therefore S$ is invertible.

$$\therefore \exists S^{-1} \text{ such that } SS^{-1} = I \\ \text{and } S^{-1}S = I$$

$$\text{Now, } TS = S^{-1}S = I$$

$$\therefore T = S^{-1}$$

$$\therefore S^{-1}S = TS = I$$

$$\text{and } SS^{-1} = ST = I$$

$$\boxed{ST = I \iff TS = I}$$

4) $S \in L(V)$.

To prove: The following are equivalent.

a) S is an isometry

~~b) $\langle Su, Sv \rangle = \langle u, v \rangle$~~

b) $\langle Su, Sv \rangle = \langle u, v \rangle \quad \forall u, v \in V$.

c) $Se_1, \dots, Sen$ is orthonormal vectors $e_1, \dots, en \in V$.

d) $\exists$ orthonormal basis $e_1, \dots, en$ of V , $Se_1, \dots, Sen$ is orthonormal.

e) $S^*S = I$

f) $SS^* = I$

g) S^* is an isometry.

h) S is invertible $\& S^{-1} = S^*$.

Proof:

for $a \Leftrightarrow b$,

$$4 \langle u, v \rangle = \|u+v\|^2 - \|u-v\|^2$$

- from Q3.

If S preserves norms,

$$\|S u\| = \|u\|$$

$$\|S u + S v\| = \|u+v\| \quad \& \quad \|S u - S v\| = \|u-v\|$$

$$\therefore \langle S u, S v \rangle$$

$$\begin{aligned} \therefore 4 \langle u, v \rangle &= \|u+v\|^2 - \|u-v\|^2 \\ &= \|S u + S v\|^2 - \|S u - S v\|^2 \\ &= 4 \langle S u, S v \rangle \end{aligned}$$

$$\boxed{\langle u, v \rangle = \langle S u, S v \rangle}$$

for reverse proof,

Let $x = u$ and $u = v$,

for $b \Rightarrow c$,

as the inner product space remains preserved ~~via~~,

- i). orthogonality is preserved.
- ii). norms are preserved.

$\therefore$ orthogonal lists remain orthogonal,

for $c \Rightarrow d$,

if it works for orthonormal lists, it has to trivially work for some $e_1, \dots, e_n$

being an ~~orthogonal~~ ^{orthonormal} basis.

$(d) \Rightarrow (b)$.

Let $e_1, \dots, e_n$ be ~~orthogonal~~ ^{normal} basis such that e_i is orthonormal.

$$u = \sum a_i e_i, v = \sum b_i e_i.$$

$$\therefore S u = \sum a_i S e_i$$

$$S v = \sum b_i S e_i$$

$$\langle S u, S v \rangle = \sum a_i \bar{b}_i = \langle u, v \rangle,$$

$$b \Leftrightarrow e.$$

$$\begin{aligned} \langle S u, S v \rangle &= \langle u, S^* S v \rangle \\ &= \langle u, v \rangle \quad \forall u, v \in V. \end{aligned}$$

$$\therefore S^* S v = v$$

$$\boxed{S^* S = I.}$$

$$e \Rightarrow f.$$

$$\text{if } S^* S = I,$$

as S is a one-one function,
its inverse is $\dim(S)$ in $\mathbb{R}^n$.

$$\therefore S^{-1} = S^*$$

$$\therefore S S^* = I.$$

and vice versa for $f \Rightarrow e$.

$$e \Rightarrow g.$$

$$(\cancel{S^*})^2$$

$$\cancel{SS^*} = I.$$

$$\therefore \cancel{S^*}$$

$$S^* S = I$$

$\therefore \cancel{S^*}$ replacing $S^* \rightarrow S'$

$$(\cancel{S^*})^* S = I$$

$$\therefore SS^* = I.$$

$\therefore$ it's an isometry.

$$e \Rightarrow h.$$

$$S^* S = I \quad \& \quad S S^* = I$$

$$\therefore S^* = S^{-1}.$$

$\therefore S$ is invertible S

$$S^* = S^{-1}.$$

$$h \gg e.$$

$$S^T = S^*$$

$$\therefore S^T S = S^* S = I.$$

$\therefore$ all the above points are equivalent.

5). V is a vector space over $\mathbb{C}$
 $T \in L(V)$ is a normal
 linear operator.

$$M(T(e_1, \dots, e_n)) = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ 0 & a_{22} & & \\ 0 & 0 & \ddots & \\ 0 & 0 & 0 & \ddots \\ \vdots & \vdots & \vdots & \vdots & \ddots \end{bmatrix}$$

if $a_{22} \neq 0$, prove that

$$a_{23} = \dots = a_{2n} = 0.$$

Proof: $TT^* = T^*T.$

T is normal $\Rightarrow$ upper $\triangleleft$ in
orthonormal basis

Now, as $\|Te_i\| = \|T^*e_i\|$,

row norm = column norm

$$\therefore \sum_k |a_{ki}|^2 = \sum_l |a_{li}|^2$$

for $i \geq 1$,

$$\sum_k |a_{ki}|^2 = |a_{1i}|^2 + |a_{2i}|^2 + \dots$$

Now, ~~only~~ for column 1,

$$\sum_k |a_{k1}|^2 = |a_{11}|^2$$

$$\therefore |a_{11}|^2 + |a_{21}|^2 + \dots = |a_{11}|^2$$

~~or~~ ~~and~~

$$|a_{21}|^2 + \dots \geq 0$$

$$\therefore a_{12} = a_{13} = \dots = a_{1n} = 0$$

Now for $i=2$,

Row $\Rightarrow$

$$|a_{22}|^2 + |a_{23}|^2 + \dots + |a_{2n}|^2$$

Now n column $\Rightarrow$ 2,

$$a_{12} = 0$$

$$\therefore |a_{22}|^2 + |a_{23}|^2 + \dots + |a_{2n}|^2 = |a_{22}|^2$$

$$\therefore |a_{23}|^2 + \dots + |a_{2n}|^2 = 0$$

$$\boxed{\therefore a_{23} = a_{24} = \dots = a_{2n} = 0}$$